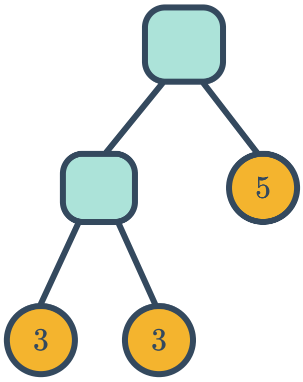
Factor trees

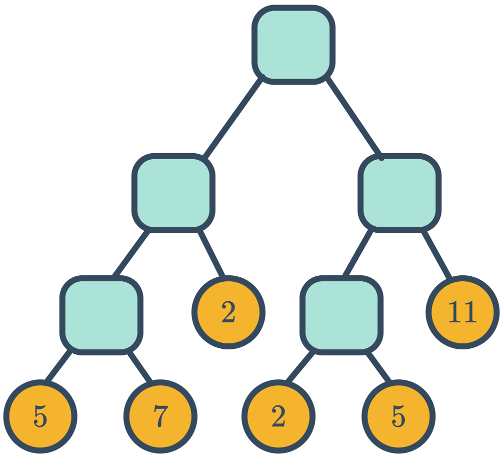
1 Select all the prime numbers from each of these lists:

a $3, 10, 12, 4, 5, 6$ b $14, 7, 15, 16, 13, 11$ c $21, 19, 27, 17, 22, 1$

2 A number has the following factor tree:
What is this number at the top of the tree?



3 A number has the following factor tree:
What is this number at the top of the tree?



4 Construct a factor tree for the following numbers:

a 6 b 12 c 28 d 42

e 81 f 144 g 200 h 360

Prime factorisations

5 Which of the following shows a correct prime factorisation in expanded form?

a $175 = 35 \times 5$

b $20 = 2 \times 2 \times 5$

c $286 = 2 \times 7 \times 11$

d $27 = 3^3$

6 Which of the following shows a correct prime factorisation in index form?

a $36 = 2 \times 2 \times 2 \times 3$

b $770 = 3^2 \times 55$

c $200 = 2^3 \times 5^2$

d $21 = 3^2 \times 7$

7 Write the following values as a product of prime factors in expanded form:

a 36 b 42 c 54 d 48

e 72 f 96 g 132 h 148

8 Write the following values as a product of prime factors in index form:

a 90 b 200 c 180 d 225

e 208 f 400 g 560 h 504

9 For the following values:

i Write as a product of prime factors in expanded form.

ii Using your answer from part (i), list all the factors.

a 12 b 35 c 20 d 40

e 52 f 54 g 66 h 275

10 List all the factors of 100.